

Ali TOURANI

COMPUTER VISION · EMBODIED AI · ROBOTICS

Strassen, Luxembourg

Phone: (+352) 661 697 868 | atourani@acm.org | Website: alitourani.github.io | GitHub: [alitourani](#) | LinkedIn: [alitourani](#) | Medium: [alitourani](#) |

Scholar: [alitourani](#)

Research Profile

Former Postdoctoral Research Associate and R&D Specialist at **SnT, University of Luxembourg**, expert in **Computer Vision**, **embodied AI**, **LLM/VLM** grounding, and **Multimodal RAG** for autonomous robot perception and situational awareness. Author of **30+ peer-reviewed publications** in top-tier venues (IEEE RA-L, IROS, ACM TORS – **h-index: 15**). Developer of **10+ open-source software frameworks** (Python, C++, ROS) backed by cutting-edge research. **10+ years** of applied research expertise spanning visual/multimodal data processing for reasoning and information retrieval, mentoring Ph.D./M.Sc. researchers, lecturing undergraduate students, and leading interdisciplinary research projects.

Education

Ph.D. in Computer Science (Doctorat en Informatique)

University of Luxembourg, Luxembourg | April 2022 – March 2026

- **Thesis:** Enhancing Robots' Situational Awareness using Imperceptible Artificial Landmarks ([thesis link](#))
- **Achievement:** Nominated for the **Excellent Thesis Award** of the University of Luxembourg (March 2026)
- **Affiliation:** Automation and Robotics Research Group, Interdisciplinary Centre for Security, Reliability, and Trust (SnT) ([link](#))
- **Supervisor:** Prof. Dr. Holger Voos ([link](#)) | **Advisor:** Dr. Jose Luis Sanchez-Lopez ([link](#))

M.Sc. Computer Engineering (Software Engineering Stream)

University of Guilan, Iran | September 2016 – January 2019

- **Thesis:** Automatic Vehicle Speeding Violation Detection using Computer Vision Solutions ([link](#))
- **Supervisor:** Prof. Asadollah Shahbahrami ([link](#))
- **GPA:** 18.97 out of 20.00 (Excellent)

B.Sc. Computer Engineering (Software Engineering Stream)

University of Guilan, Iran | September 2009 – June 2013

- **Thesis:** An E-Ticketing Software Platform for Appointment Scheduling in Medical Clinics

Academic & Industry Experience (since 2019)

Interdisciplinary Centre for Security, Reliability, and Trust (SnT), Luxembourg

Postdoctoral Research Associate and R&D Specialist (full-time) | April 2026 – July 2026

- Led research on **Vision-Language Model (VLM)** grounding for robotic scene understanding, spatial reasoning, and knowledge retrieval.
- Designed and deployed an **embodied AI** core for spatial reasoning and **context-aware perception** in unstructured environments.
- Implemented a **Retrieval-Augmented Generation (RAG)** system with end-to-end **GenAI**-driven reasoning for real-world robotics.
- Mentored **three Ph.D. students** across system development, experimental design, and publication workflows.

Doctoral Researcher (full-time) | April 2022 – March 2026

- Developed various **computer vision** and **spatial AI** frameworks to improve autonomous robots' situational awareness.
- Designed, packaged, and maintained **5+ open-source** robotic software packages, adopted widely across industry and research teams.
- Authored **15+ publications** across top-tier scientific venues (journals and conferences), achieving an **h-index of 15**.

AVID Technology Development Co., Iran

Full-Stack Software Developer (full-time) | January 2019 – December 2021

- Built React/Node.js **full-stack** systems for **international** enterprise clients across **5+ countries**, as production-grade software delivery.
- Mentored junior developers and led **Agile/Scrum** sprints as secondary Scrum Master for **teams of 10+** people.
- Implemented **CI/CD pipelines**, DevOps/MLOps workflows, and **Docker-based** deployments to support reliable production releases.

Department of Computer Engineering, University of Guilan, Iran

University Lecturer (contract) | September 2019 – September 2021

- Lectured **Computational Intelligence** and **Software Laboratory** courses over four semesters to **100+ undergraduate students**.
- Supervised **+30 student projects** focused on **machine learning**, **deep learning**, and **software design**.

Guilan Science and Technology Park Business Incubator, Iran

Research Software Engineer (part-time) | July 2019 – July 2021

- Deployed two production-grade computer vision systems for **vehicle tracking** and **license plate recognition** in real-world traffic.
- Bridged research and industry by applying academic methods to scalable **Intelligent Transportation Systems (ITS)**.

A more comprehensive overview of my professional experience is available on my [personal website](#).

Selected Publications

International Conferences

2026 [A. Tourani](#), M. Fernandez-Cortizas, S. Ejaz, D. Perez-Saura, A. Bikandi-Noya, J.L. Sanchez-Lopez, and H. Voos, “**Passage-Aware Structural Mapping for RGB-D Visual SLAM**,” *IEEE/RSJ International Conference on Robotics and Automation (ICRA’26), Workshop on Robots Meet Prior Maps*, Vienna, Austria. [PDF].

2023 [A. Tourani](#), H. Bavle, J.L. Sanchez-Lopez, R. Muñoz Salinas, and H. Voos, “**Marker-based Visual SLAM leveraging Hierarchical Representations**,” *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS’23)*, pp. 3461-3467, Detroit, Michigan, USA [DOI].

Peer-review Journals

2026 [A. Tourani](#), S. Ejaz, H. Bavle, M. Fernandez-Cortizas, J.L. Sanchez-Lopez, and H. Voos, “**vS-Graphs: Tightly Coupling Visual SLAM and 3D Scene Graphs Exploiting Hierarchical Scene Understanding**,” *IEEE Robotics and Automation Letters*, vol. 11, no. 8, pp. 9239-9246 [DOI].

2026 [A. Tourani](#), F. Nazary, and Y. Deldjoo, “**RAG-VisualRec: An Open Resource for Vision-and Text-Enhanced Retrieval-Augmented Generation in Recommendation**,” *ACM Transactions on Recommender Systems*, vol. 5, no. 1, p. 27 [DOI].

The complete list of papers and preprints is available on my [Google Scholar](#) profile (h-index: 15).

Selected Projects (open-source)

Sentra GenAI Toolkit (<https://github.com/snt-arg/sentra>)

- An Embodied AI framework employing VLM grounding for real-time Multimodal RAG and spatial Visual Question-Answering.

Visual S-Graphs (vS-Graphs - https://github.com/snt-arg/visual_sgraphs)

- A robotic and spatial AI framework for semantic scene understanding and generating environments’ interpretable 3D digital twins.

RAG-VisualRec Advanced Retrieval System (<https://github.com/recsys-lab/rag-visualrec>)

- A Generative AI-based resource for enhancing reasoning with visual, audio, and textual embeddings.

Popcorn Video Recommendation Framework (<https://github.com/recsys-lab/popcorn>)

- Multimodal framework for visual embeddings extraction, fusion, augmentation, and RAG-based retrieval.

Honors & Awards

2026 **Nomination**, Best Ph.D. Thesis Award Nominee, University of Luxembourg, Luxembourg

2022 **Scholarship**, Competitive Fully Funded Ph.D. Position (TRANSCEND Project), University of Luxembourg, Luxembourg

2019 **Award**, Best Student Researcher for Scientific and Practical Contributions (class of 2016), University of Guilan, Iran

2019 **Award**, Academic Excellence: Highest GPA in M.Sc. Computer Engineering (class of 2016), University of Guilan, Iran

Skills

Software Development

Desktop Stack	Expert: Python, C++ Proficient: C# Familiar: Java (Spring), Objective-C
Web Stack	Expert: React.js, Node.js, JavaScript, CSS, HTML Proficient: Next.js, GraphQL, PHP Familiar: Webpack
Databases	Expert: Microsoft SQL, MySQL Proficient: PostgreSQL, MongoDB, SQLite Familiar: Elasticsearch, Neo4j
Extras	Expert: Git, Linux, Jira Proficient: Docker, REST API, .NET Framework Familiar: CI/CD, VMware

Tools & Technologies

Computer Vision	Expert: OpenCV, YOLO Proficient: CLIP, Scikit-Image Familiar: EmguCV, AForge.NET
Machine Learning	Expert: HuggingFace Transformers, Keras Proficient: PyTorch, TensorFlow Familiar: Scikit-learn
Robotics	Expert: ROS2, RViz Proficient: PCL, Voxblox Familiar: GTSAM, Gazebo
Generative AI & LLM	Expert: VLMs, VLAs Proficient: RAG, GPT Familiar: Llama, SentenceTransformers

Soft Skills

Communication	Presentation, Scientific Writing, Technical Documentation
Management	Agile, Scrum, Mentoring (students, developers, interns), Project Coordination
Teamwork	Interdisciplinary collaboration, Team spirit, Cross-cultural collaboration
Languages	English (C1), French (B1), German (A2), Luxembourgish (A1), Turkish (A1), Persian (mother tongue)

Reference

Prof. Dr. Holger Voos (holger.voos@uni.lu)

Full Professor in Engineering Science, University of Luxembourg, Luxembourg (Ph.D. Supervisor)

Dr. Jose Luis Sanchez-Lopez (joseluis.sanchezlopez@uni.lu)

Senior Research Scientist, Automation & Robotics Group (ARG), SnT, Luxembourg (Ph.D. Advisor)

Dr. Yashar Deldjoo (deldjooy@acm.org)

Senior Research Scientist and Associate Professor, Polytechnic University of Bari, Italy (Research Collaboration)